

Linear Circuits 2

Circuit Design #2 Lab Report

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Objective

The goal of this project is to use our knowledge of the breadboard and circuit analysis to produce a physical interpretation of the example circuits that follow certain constraints. For each part there were different instructions.

Part A

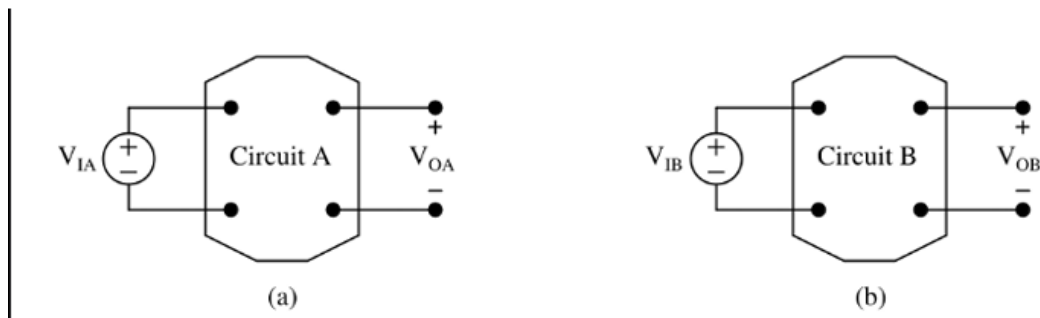


Figure 1: Circuit Design Part A

For this part, in circuit A, the constraint is $V_{OA} = K_a \cdot V_{IA}$, and circuit B's was $V_{OB} = K_b \cdot V_{IB}$. $K_a = 0.75$ and $K_b = 0.6$. While doing this we needed resistors between $1\text{ k}\Omega$ to $10\text{ k}\Omega$ and had to use the fewest components possible.

Part B

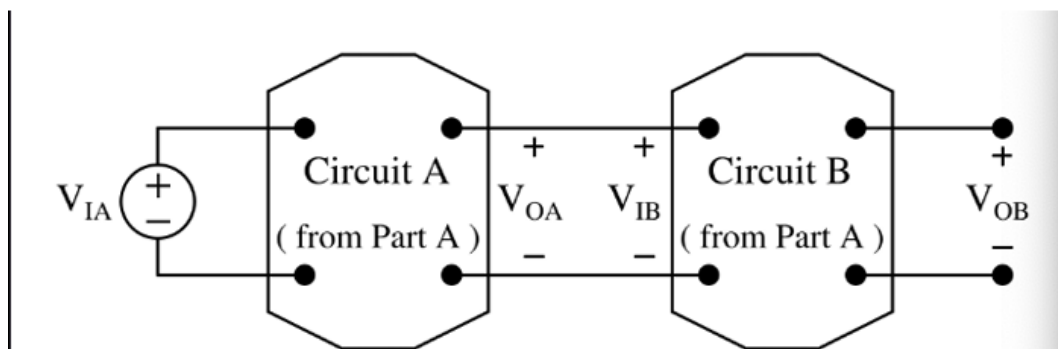


Figure 2: Circuit Design Part B

For the second part of this project, we are now asked to cascade circuit A and B from the previous portion. Using this, we figure out whether $V_{OB} = K_a \cdot K_b \cdot V_{IA}$ and explain why it does not.

Following this, we need to redesign one of the two circuits to not only satisfy that requirement but also the constraints from before, $V_{OA} = K_a \cdot V_{IA}$ and $V_{OB} = K_b \cdot V_{IB}$. Also keeping in mind to use the fewest components possible and only resistances between $100\ \Omega$ and $400\ \text{k}\Omega$.

Part C

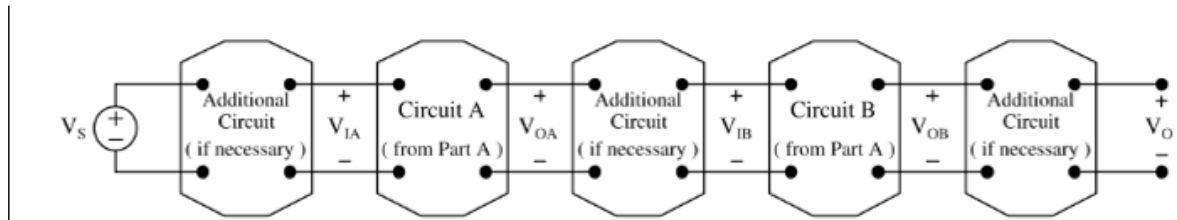


Figure 3: Circuit Design Part C

Lastly, for Part C you will use Circuit A and B from Part A to design a circuit that will exhibit the same behavior as Figure 3. Using additional circuits if needed, an op-amp and resistors from $1\ \text{k}\Omega$ to $10\ \text{k}\Omega$. The following constraints must be satisfied: $V_{OA} = K_a \cdot V_{IA}$, $V_{OB} = K_b \cdot V_{IB}$, and $V_O = K_a \cdot K_b \cdot K_c \cdot V_S$. $K_c = 8$.

Equipment

- Lab Equipment
- LTspice
- Hantek 3-in-1 Digital Equipment
- Powered Breadboard
- $1\ \text{k}$, $3\ \text{k}$, $4\ \text{k}$, $4.3\ \text{k}$, $6\ \text{k}\ \Omega$ resistors
- Operational Amplifier

Simulations / Hand Calculations

Part A

For Part A, we needed to design a circuit that could satisfy $V_{OA} = K_a \cdot V_{IA}$, and another circuit that satisfies $V_{OB} = K_b \cdot V_{IB}$, with values $K_a = 0.75$ and $K_b = 0.6$.

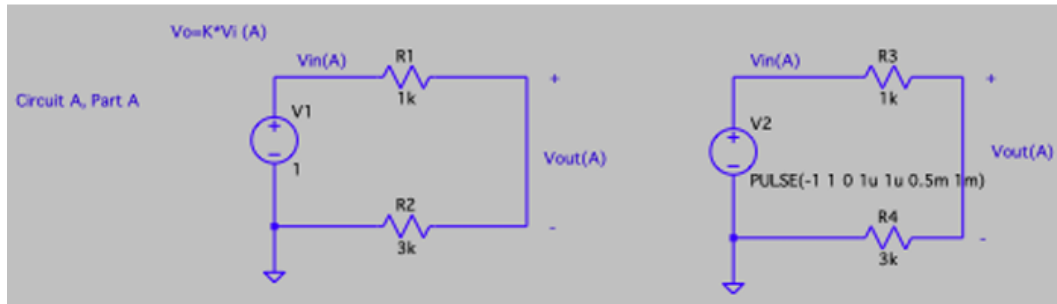


Figure 4: Circuit (a) in Part A

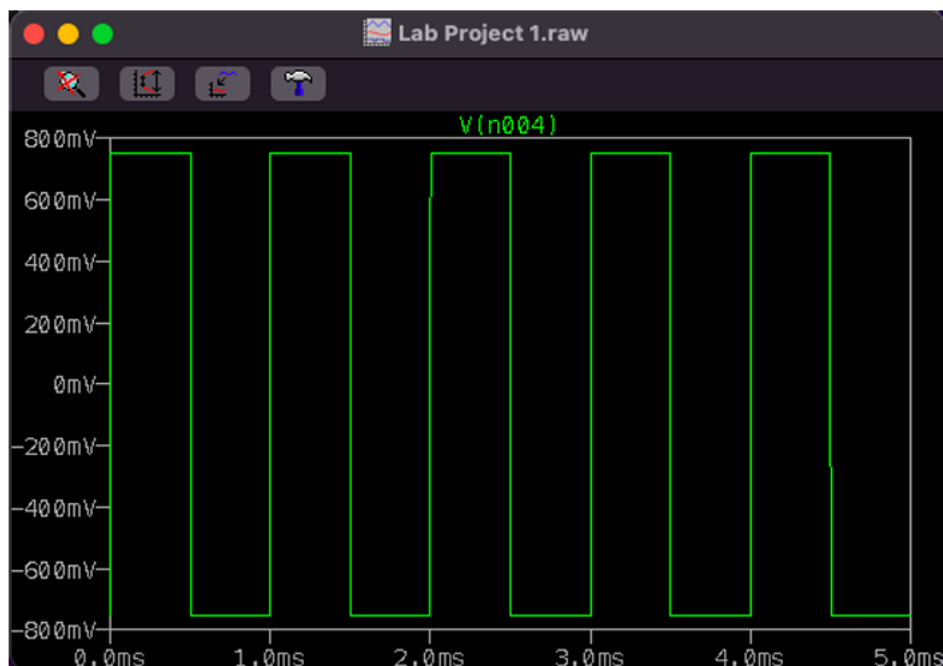
Figure 5: $V_{OA}(0.75V) = V_{in}(1V) \cdot K_a(0.75)$ 

Figure 6: Waveform for square-wave voltage

For Circuit A, we determined the resistor values $1\text{ k}\Omega$ and $3\text{ k}\Omega$ would satisfy the requirements.

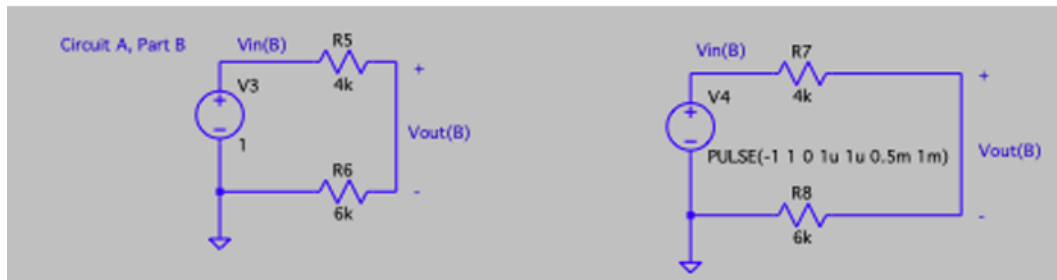


Figure 7: Circuit (b) in Part A

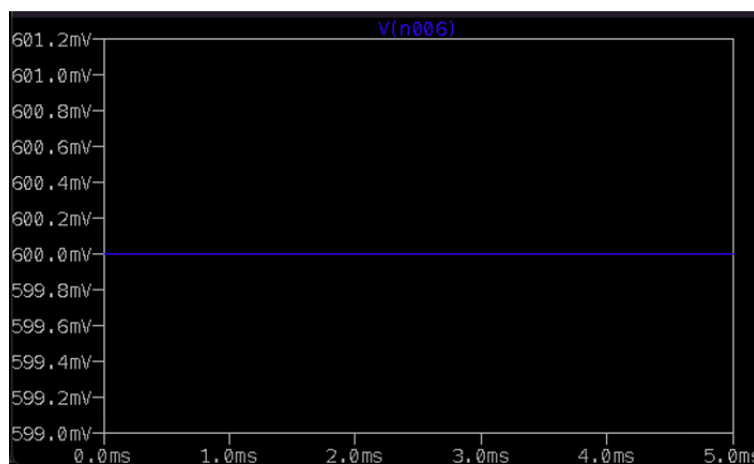


Figure 8: $V_{OB}(0.6\text{V}) = K_b(0.6) \cdot V_{in}(1\text{V})$

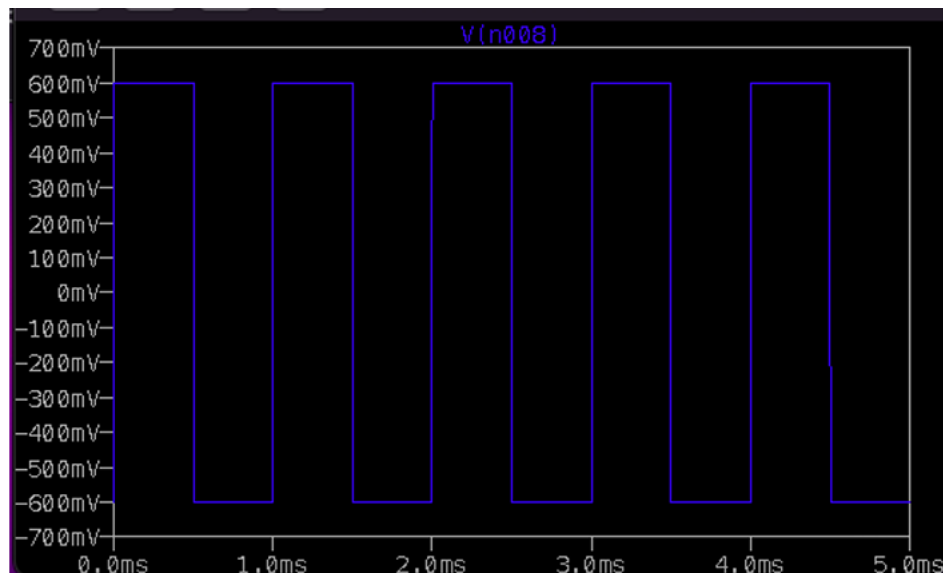


Figure 9: Waveform for square-wave voltage

For Circuit B, we determined that $4\text{ k}\Omega$ and $6\text{ k}\Omega$ were good resistor values to yield the proper results.

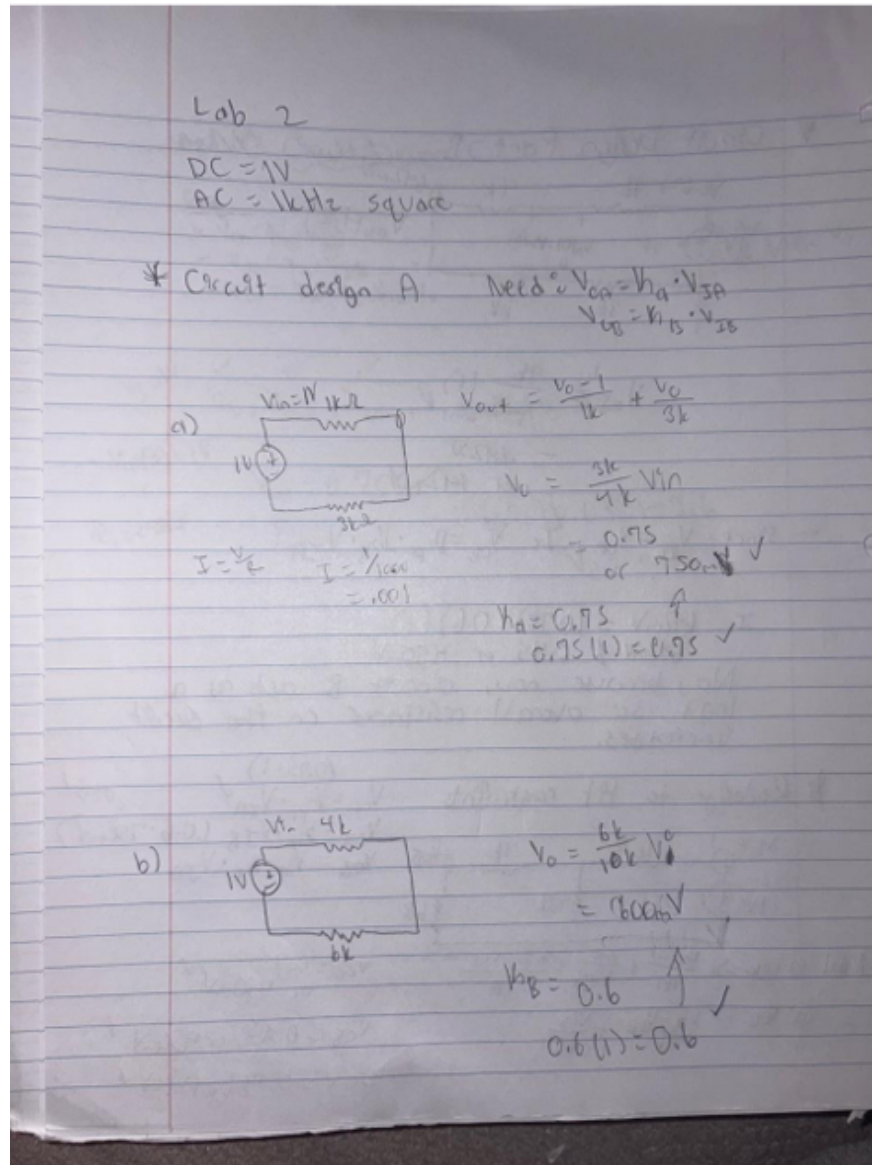


Figure 10: Hand calculations for both circuits in Part A

Part B

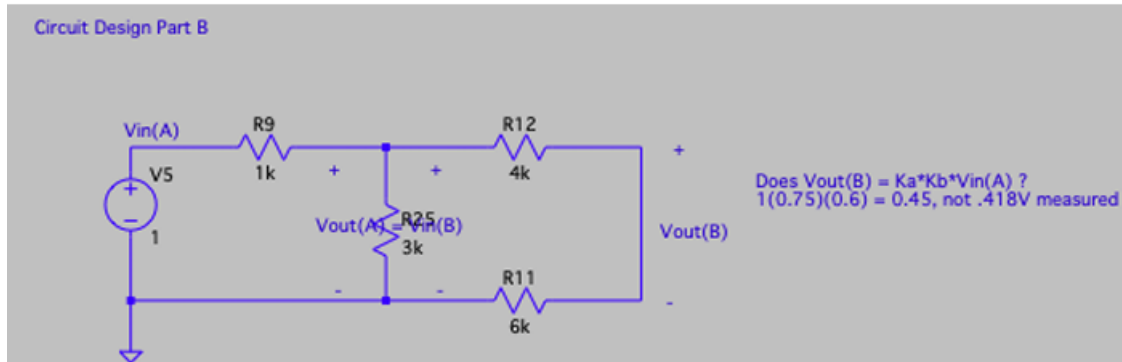


Figure 11: Cascaded original circuit outputs 418 mV, so $V_{OB} = K_a \cdot K_b \cdot V_{IA}$ does not apply

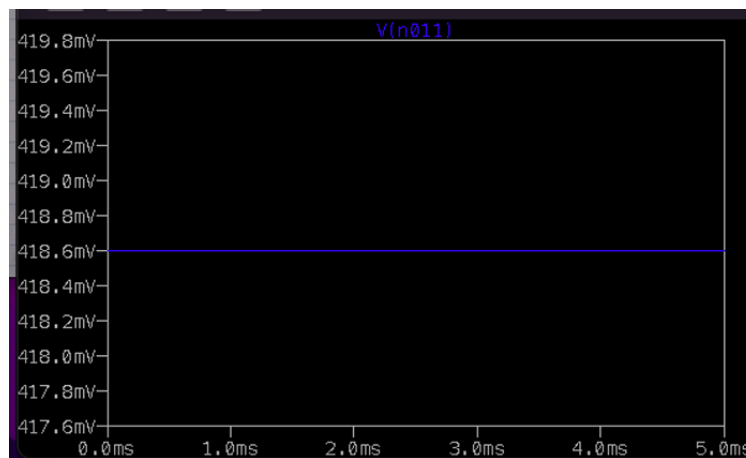


Figure 12: $V_{OUT(B)} = 418.6 \text{ mV}$

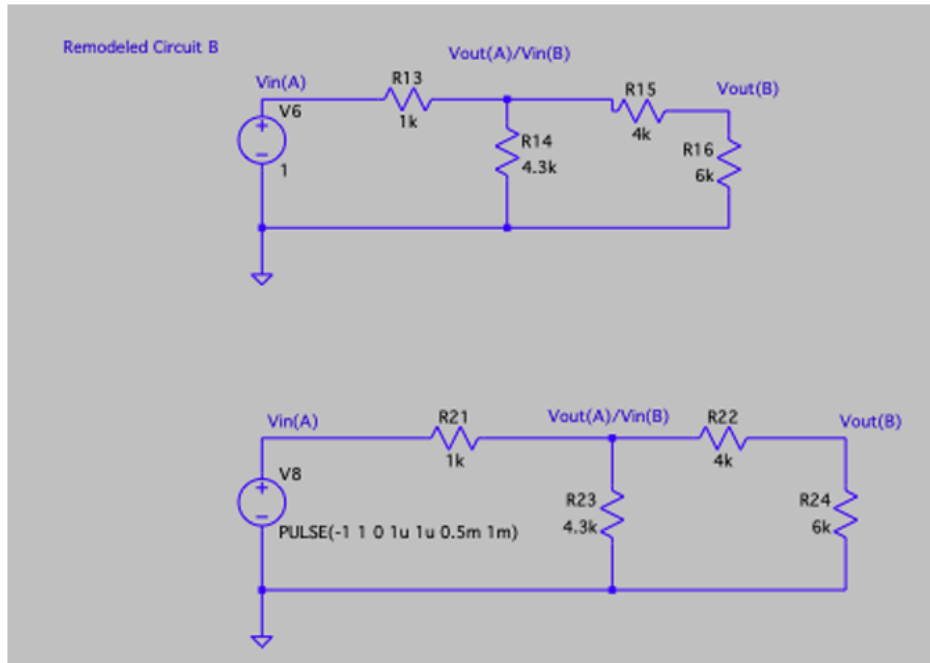


Figure 13: Remodeled version of Circuit B ($3\text{ k}\Omega$ changed to $4.3\text{ k}\Omega$)

After concluding the original circuit resulted in a voltage output that didn't align with the equation, we modified circuit (a) and changed the $3\text{ k}\Omega$ resistor to a $4.3\text{ k}\Omega$ one. As shown below, it reflects 450 mV , which is correct for the given constraint.



Figure 14: $V_{OB}(450\text{mV}) = V_{IA}(1\text{V}) \cdot K_a(0.75) \cdot K_b(0.6)$

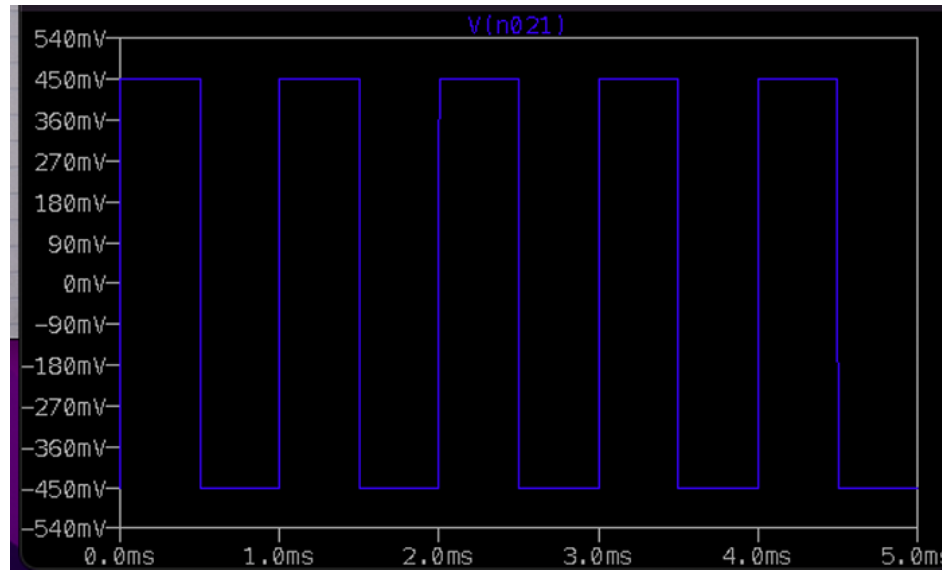


Figure 15: Voltage waveform for square wave

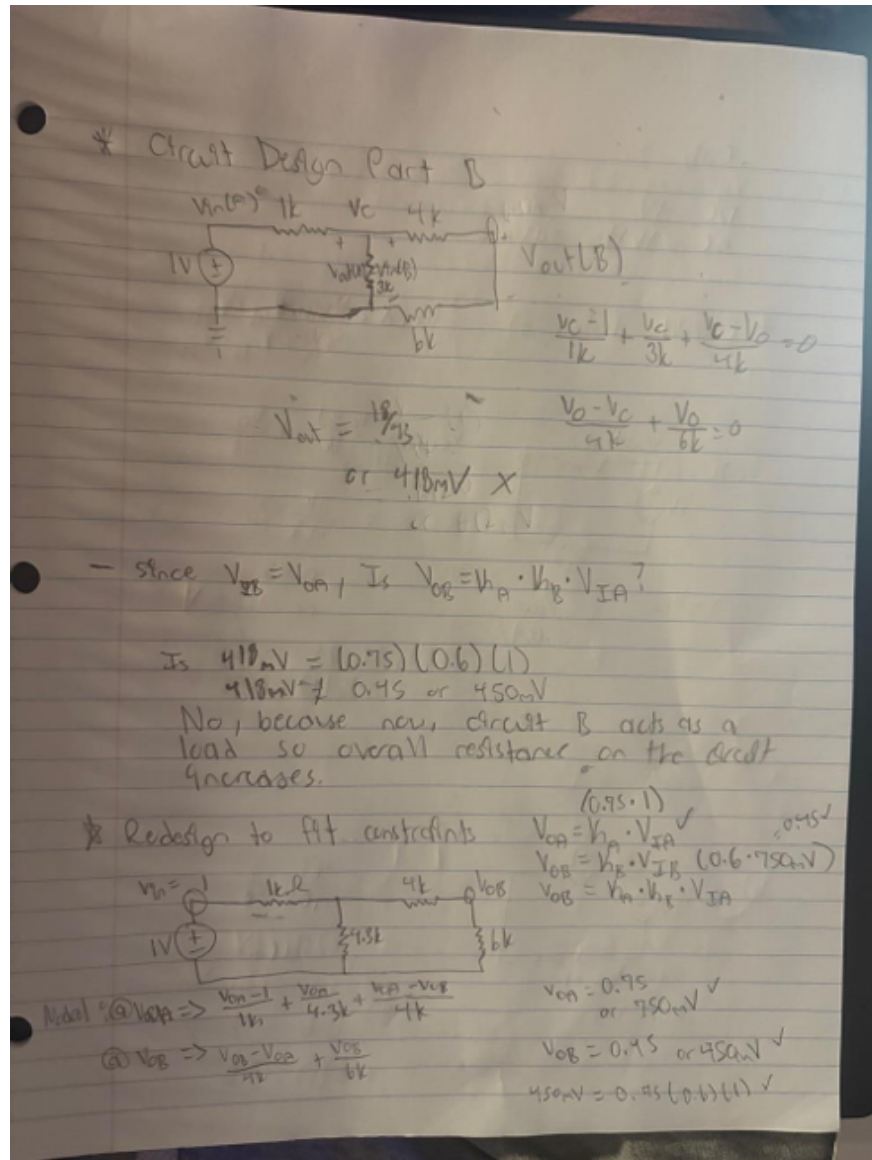


Figure 16: Hand calculations for Part B

Since $V_{OB} = V_{OA}$, is $V_{OB} = K_a \cdot K_b \cdot V_{IA}$? Checking: $418 \text{ mV} \stackrel{?}{=} (0.75)(0.6)(1 \text{ V}) = 450 \text{ mV}$. No, because the new Circuit B acts as a load, so the overall resistance in the circuit increases. Redesigning to fit the constraints (changing the $3 \text{ k}\Omega$ resistor to $4.3 \text{ k}\Omega$) confirms $V_{OA} = 0.75 \text{ V}$ (750 mV) and $V_{OB} = 0.45 \text{ V}$ (450 mV), matching $V_{OB} = K_a \cdot K_b \cdot V_{IA}$.

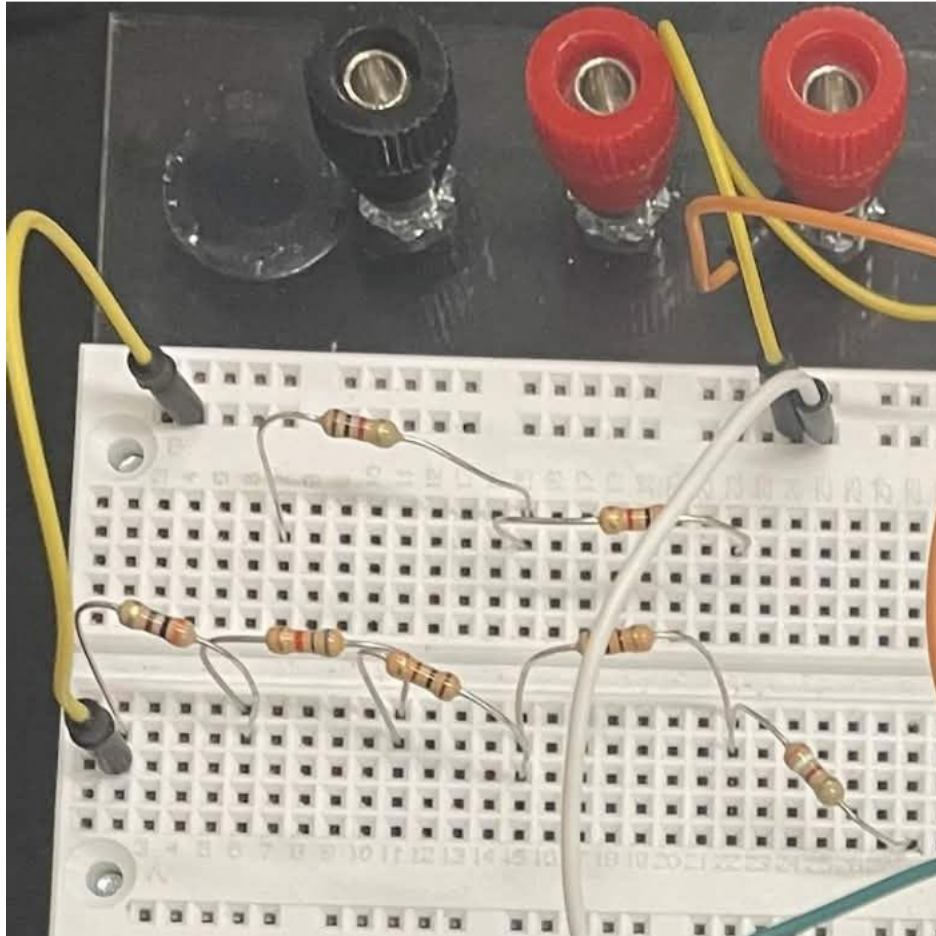


Figure 18: Breadboard implementation of Circuit A

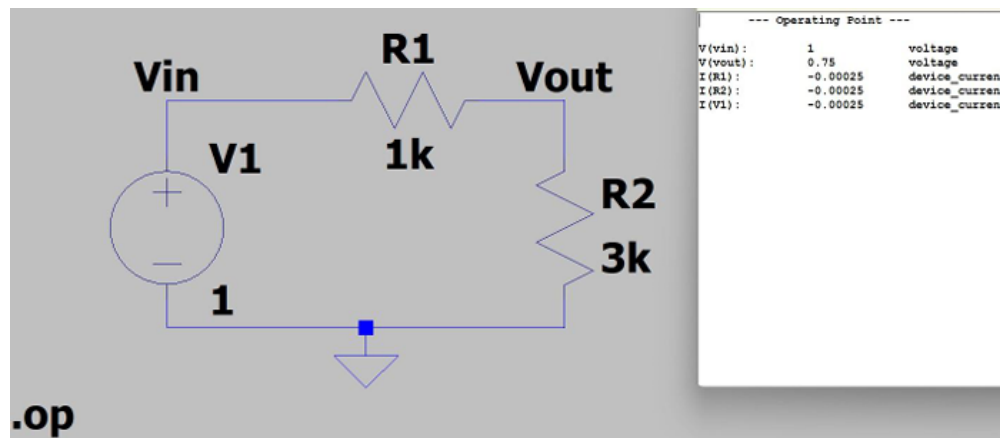
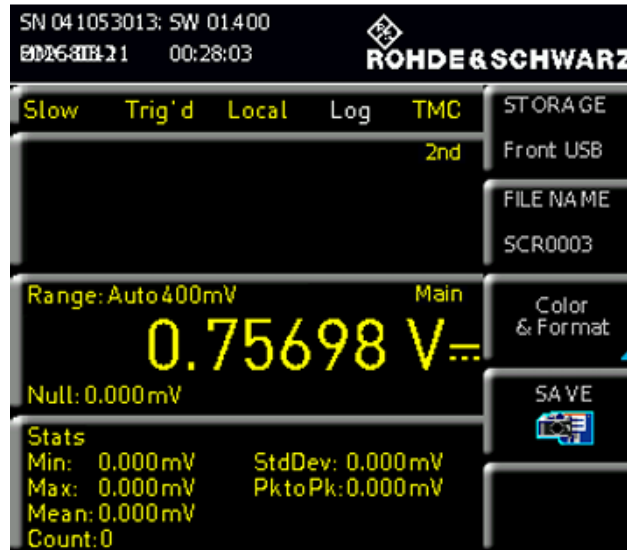


Figure 19: Circuit A schematic with simulated operating-point data

Figure 20: Measured $V_{OA} = 0.75698 \text{ V}$

Measurement of V_{OB} of circuit B = 0.60021 V:

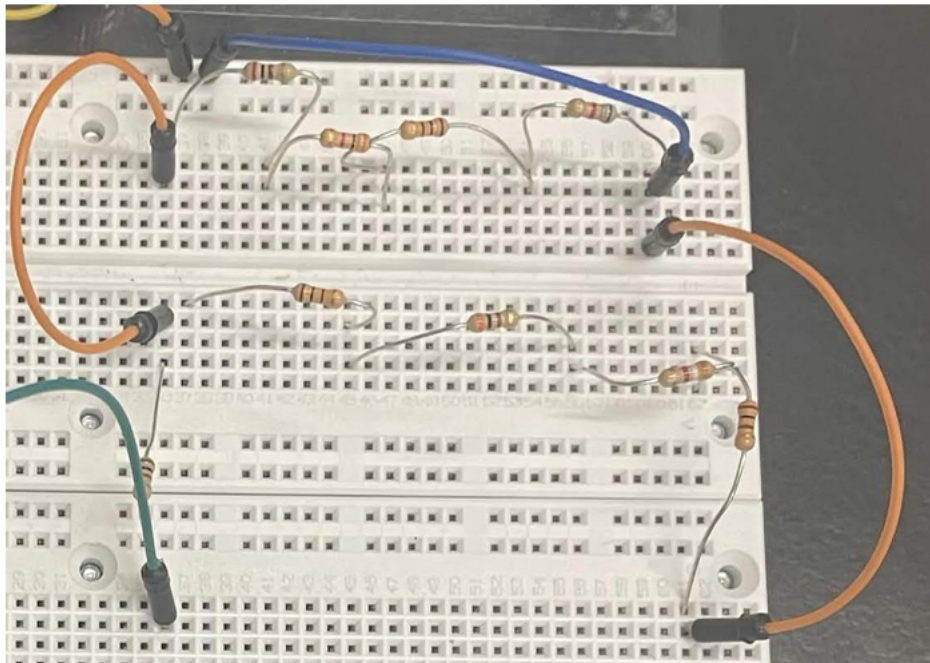


Figure 21: Breadboard implementation of Circuit B

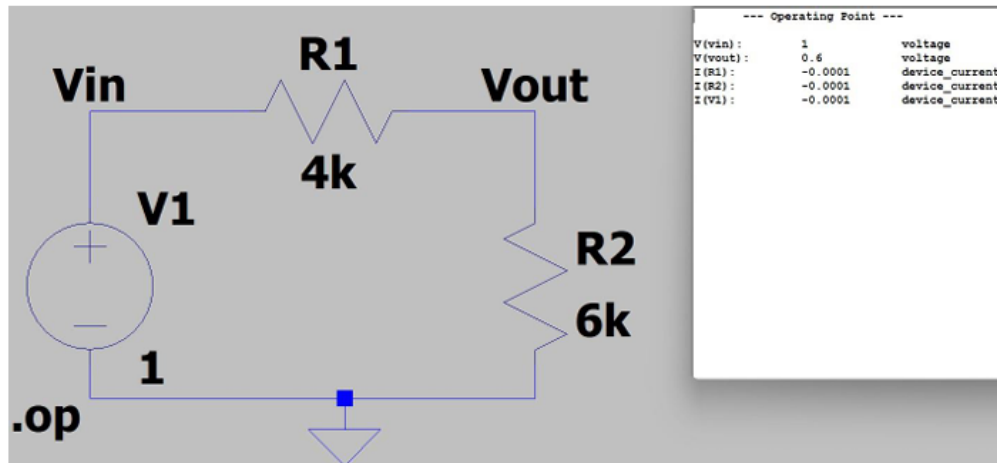
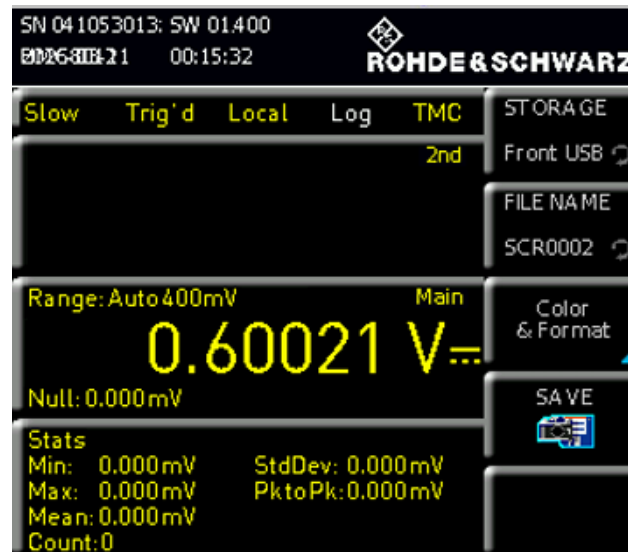


Figure 22: Circuit B schematic with simulated operating-point data

Figure 23: Measured $V_{OB} = 0.60021$ V

The node of $V_{OUT(B)}$ of the noninverting op-amp:

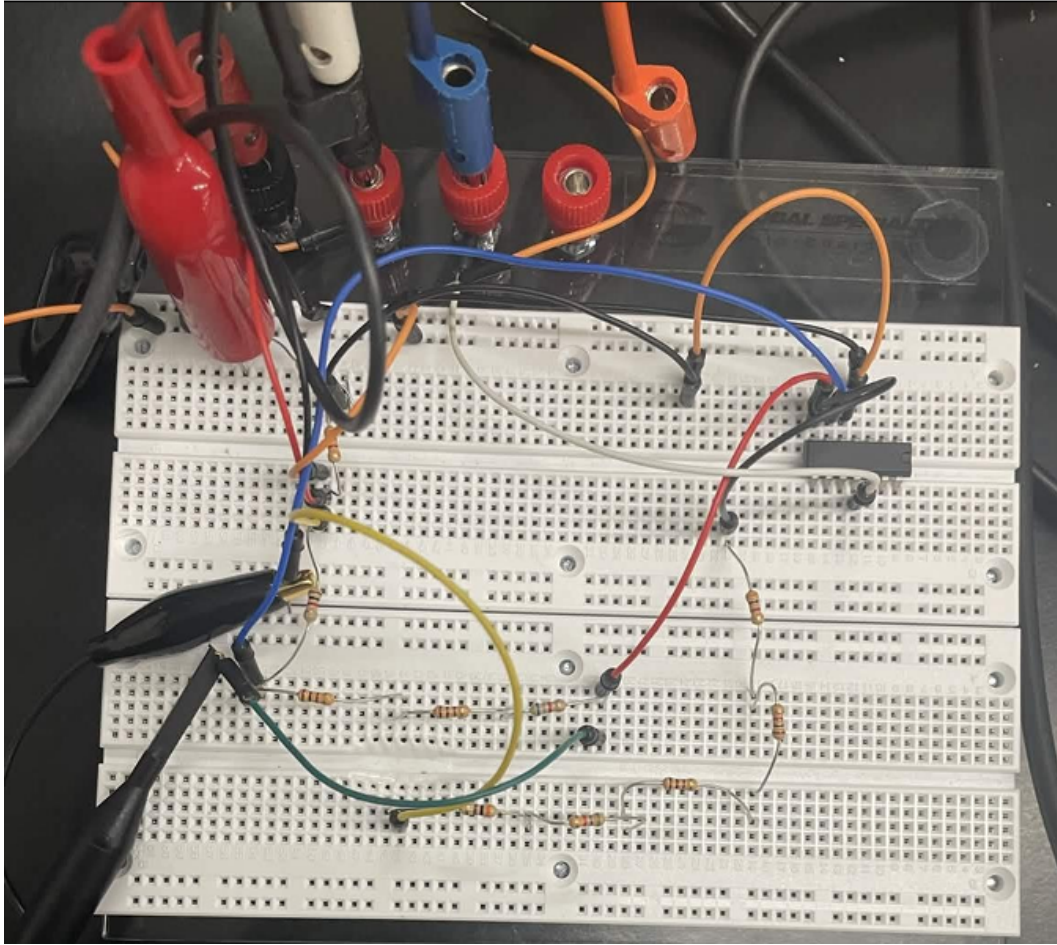


Figure 24: Breadboard implementation of the full Part C cascaded circuit with op-amp

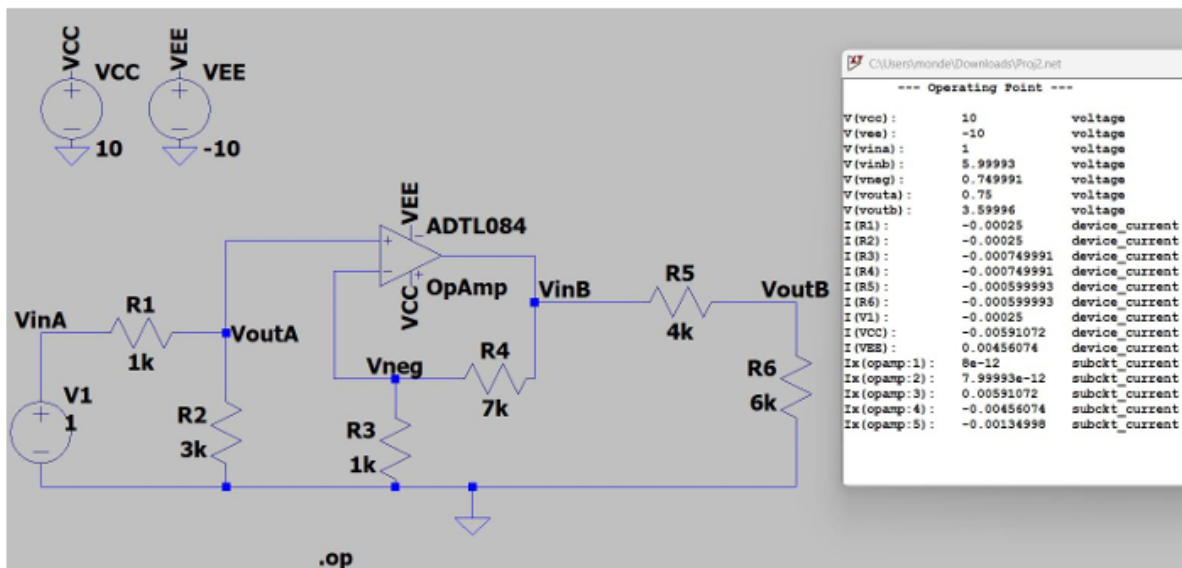


Figure 25: Part C op-amp circuit with simulated operating-point data (repeated for reference)

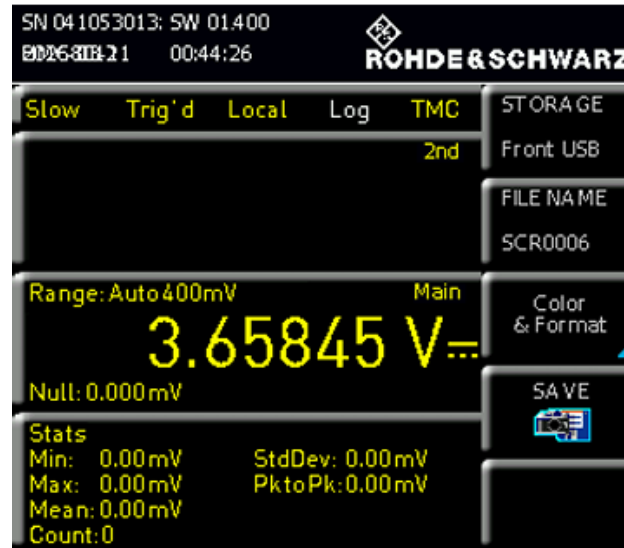


Figure 26: Measured output of the gain-of-8 op-amp stage, 3.65845 V

Results

Across the different parts of the experiment, we observe that the measured results closely match the simulated values, confirming the designs to be true. In Part A, the measured output of 0.757 V is closely accurate with the simulated value of 0.75 V. In Part B, the measured output of 0.60 V is accurate with the simulation result. Finally, in Part C, the op-amp circuit produced an output of 3.658 V, which is in range of the simulated output of 3.6 V. Overall, the experimental and simulated results verify the theoretical designs with negligible errors.

Conclusion

This experiment has shown how unrelated circuit stages can be combined to obtain a certain overall gain. In Part A and Part B, we designed voltage divider circuits with gains of $K_a = 0.75$ and $K_b = 0.6$. We then cascaded the circuits to observe how the individual stages would affect the total gain. Noticing the differences in the gain at certain points, we determined that we must do a resistor redesign to regain the gains we originally had. In Part C, we designed an op-amp circuit with a gain factor of $K_c = 8$, which restores ideal behavior while allowing for additional amplification of voltage. Overall, the experiment reinforced how theoretical gain relationships combined with careful cascaded redesigns could be achieved accurately under real-world conditions.